



2 The curve C has parametric equations

$$x = \sinh t \quad \text{and} \quad y = t + \cosh t.$$

(a) Find $\frac{dy}{dx}$ in terms of t . [2]

[illegible]

(b) Show that $\frac{d^2y}{dx^2} = \frac{1 - \sinh t}{\cosh^3 t}$. [4]

[illegible]



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(c) Find the Maclaurin's series for y in terms of x up to and including the term in x^2 . [2]

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